DOINGG@UNION.EDU

DOING-LAB-NOTEBOOK

SELF

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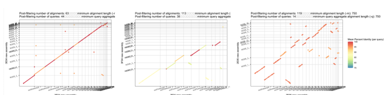

The shared electronic lab notebook of the Doing Com-Bio Lab.

Announcements

Something

Something else

Schedule

Weekly Schedule

Schedule

check often as things may change

W	TOPIC	Reading	Due
1	What is bioinformatics: what are computers and living cells good at?	Bioinformatics for Dummies (BFD) Ch. 1 & 2	Survey, Quiz 0
2	Algorithms & the Genetic Code	Think Python (TP) Ch. 1, 2, 3.4–3.7 & Molecular & Cell Biology (MCFD) Ch. 1, 6, 7	HW 1, Quiz 1
3	Modular Programming & Proteins	TP Ch. 5.1–5.7, 5.11, 6.1 & MCFD Ch. 16, 17	HW 2, Quiz 2
4	Loops & Sequencing	TP Ch. 6.1–6.1, 7.1–7.3, 7.7 & MCFD Ch. 19	HW 3, Quiz 3
5	Sequence Processing & Disease	BFD Ch. 7	HW 4, Quiz 4
6	Sequence Alignment	BFD Ch. 9, TP Ch. 8	HW 5, Quiz 5
7	Multiple Sequence Alignment	BFD Ch. 13	Project 1 Outline

W	TOPIC	Reading	Due
8	Polymerase Chain Reaction	BDF Ch. 13, 9 cont.	Project 1
9	Phylogenetics	BFD Ch. 13 cont.	Project 2 Check-in
10	Gene Prediction	BFD Ch. 5, pp. 145-153	Project 2

Assignments

HW 01

CSC/BIO 243: Assignment 1

- Remember to check the schedule for readings: https://georgiadoing.github.io/CSC243-S26/02-Main/schedule.html
- Over the weekend
  - finish reading Bioinformatics for Dummies Ch. 1 & 2 and
  - begin the reading Think Python Ch. 1-3 or Molecular & Cell Biology for Dummies Ch. 1,6 and 7
  - you will want to be caught up on the reading since we will have our **first quiz next Wed of Fri**
  - I can answer any questions about materials from readings on Monday, ahead of the quiz

Part 1

- Intro Survey
- Read the Therac25 case study summary and one more detailed article (linked in week1 notes, you can choose one of the two longer articles on the notes page or find another if you prefer. Read critically and carefully and come to class **Wednesday 4/1/26** ready to discuss.

Part 2

- Prepare a 5 minute introduction to a topic for which you bring background knowledge to this class
  - you can use slides, write on the board and/or have me post/print out

handouts

- Each topic will be presented at the beginning of the relevent class day: Topic Sign-up and Schedule
- Based on the results on the survey, I have done my best to match students to topics they are already familiar with, HOWEVER, you should MOVE or ADD your name so you are comfortable with your topic
- Up to 2 students can sign up for each topic
- Questions that arise as you are preparing your topic are great reasons to come to office hours, I am happy to work with you!
- **This presentation will earn you one quiz grade, "Quiz 0" (5% of your final grade)**

Weekly Notes

Week 1

Notes from Week 1

For Reading/HW

- Therac25 Summary
- Therac25 Leveson
- Therac25 Silvis

Monday in class

- No More DNA (0.5 pts)

Wednesday in class

- Slides
- What are computers good at? (0.5 pts)

Friday in class

- Anatomy of a function
- Logging on to lab computers
- Pair Programming
- Rosalind Problems for Learning Python
- Rosalind Bioinformatics Practice Problems
- Rosalind Bioinformatics Textbook Problems

About

References